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SQL MURDER MYSTERY: WHO KILLED THE CEO?

Every clue tells a story



Data never lies — people do

Investigation by: Gagandeep Kaur Bhatti (Data Analyst)

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★ *Capstone Project — SQL Murder Mystery Investigation*

Who Killed the CEO of TechNova Inc.?

🔍 What This Project Covers

- Full 21-Day SQL Skill Application
- Use of Joins, Subqueries, CTEs & Aggregations
- Evidence Analysis (Keycards, Calls, Alibis, Locations)
- Logical Investigation & Final Case Conclusion

Special Thanks

Indian Data Club — for hosting the 21-Day SQL Challenge

DPD ZERO — for sponsoring and supporting this learning journey



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▣ **CASE FILE: CEO MURDER INVESTIGATION**

Case Title: SQL Murder Mystery - The Death of TechNova's CEO

Case ID: TN-OCT15-2025-09PM

Date & Time of Incident: October 15, 2025 — 9:00 PM

Location: CEO's Office, TechNova Inc. Headquarters



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Incident Summary

At approximately 9:00pm, the CEO of TechNova Inc. was discovered dead inside their private office.

No witnesses were present at the time of the incident.

The crime scene shows no signs of forced entry, suggesting the killer may have had authorized access.



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 Assigned Investigator

Investigator Name: Gagandeep Kaur Bhatti

Role: Lead Data Analyst

Mission: Solve the murder using SQL-based digital forensics.

All critical evidence exists within the organization's systems and databases.

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Primary Data Sources (Clue Databases)

employees
employee_id INT
name VARCHAR(50)
department VARCHAR(50)
role VARCHAR(50)
Indexes

keycard_logs
log_id INT
employee_id INT
room VARCHAR(50)
entry_time TIMESTAMP
exit_time TIMESTAMP
Indexes

calls
call_id INT
caller_id INT
receiver_id INT
call_time TIMESTAMP
duration_sec INT
Indexes

alibis
alibi_id INT
employee_id INT
claimed_location VARCHAR(50)
claim_time TIMESTAMP
Indexes

evidence
evidence_id INT
room VARCHAR(50)
description VARCHAR(255)
found_time TIMESTAMP
Indexes

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PROLOGUE: Understanding the Dataset

Pre-Investigation Queries (Data Familiarization)

Employees

employee_id	name	department	role
1	Alice Johnson	Engineering	Software Engineer
2	Bob Smith	HR	HR Manager
3	Clara Lee	Finance	Accountant
4	David Kumar	Engineering	DevOps Engineer
5	Eva Brown	Marketing	Marketing Lead
6	Frank Li	Engineering	QA Engineer
7	Grace Tan	Finance	CFO
8	Henry Wu	Engineering	CTO
9	Isla Patel	Support	Customer Support
10	Jack Chen	HR	Recruiter

Clue Provided:

✓ Helps identify who the suspects are – their names, departments, and roles.

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keycard_logs

log_id	employee_id	room	entry_time	exit_time
1	1	Office	2025-10-15 08:00:00	2025-10-15 12:00:00
2	2	HR Office	2025-10-15 08:30:00	2025-10-15 17:00:00
3	3	Finance Office	2025-10-15 08:45:00	2025-10-15 12:30:00
4	4	Server Room	2025-10-15 08:50:00	2025-10-15 09:10:00
5	5	Marketing Office	2025-10-15 09:00:00	2025-10-15 17:30:00
6	6	Office	2025-10-15 08:30:00	2025-10-15 12:30:00
7	7	Finance Office	2025-10-15 08:00:00	2025-10-15 18:00:00
8	8	Server Room	2025-10-15 08:40:00	2025-10-15 09:05:00
9	9	Support Office	2025-10-15 08:30:00	2025-10-15 16:30:00
10	10	HR Office	2025-10-15 09:00:00	2025-10-15 17:00:00
11	4	CEO Office	2025-10-15 20:50:00	2025-10-15 21:00:00

Clue Provided:

- ✓ Shows who entered or exited the CEO's office around the time of the murder or on 15 Oct.
- ✓ Helps build a timeline of movements inside the building.
- ✓ Reveals suspicious entries that do not match claimed alibis.

Calls

call_id	caller_id	receiver_id	call_time	duration_sec
1	4	1	2025-10-15 20:55:00	45
2	5	1	2025-10-15 19:30:00	120
3	3	7	2025-10-15 14:00:00	60
4	2	10	2025-10-15 16:30:00	30
5	4	7	2025-10-15 20:40:00	90

Clue Provided:

- ✓ Highlights suspicious phone calls made shortly before the crime.
- ✓ Shows communication between employees involved or coordinating.
- ✓ Helps detect panic calls, last-minute instructions, or secret conversations.

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Alibis

alibi_id	employee_id	claimed_location	claim_time
1	1	Office	2025-10-15 20:50:00
2	4	Server Room	2025-10-15 20:50:00
3	5	Marketing Office	2025-10-15 20:50:00
4	6	Office	2025-10-15 20:50:00

Clue Provided:

- ✓ Reveals who claimed to be where at the time of the murder.
- ✓ Helps catch employees who lied about their location.

Evidence

evidence_id	room	description	found_time
1	CEO Office	Fingerprint on desk	2025-10-15 21:05:00
2	CEO Office	Keycard swipe logs mismatch	2025-10-15 21:10:00
3	Server Room	Unusual access pattern	2025-10-15 21:15:00

Clue Provided:

- ✓ Lists items found at the crime scene or nearby rooms.
- ✓ Can be matched with employee movements.

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DATASET SUMMARY

After running the prologue queries, I have found:

Employees:

- 10 employees across 5 departments

Keycard Logs (Oct 15 only):

- 11 total entries
- Most activity: 8:00 AM - 6:00 PM (normal business hours)
- One suspicious late entry: 8:50 PM - 9:00 PM to CEO Office

Calls:

- 5 calls throughout the day
- 2 calls in the evening (8:40 PM and 8:55 PM)

Evidence:

- 3 pieces of evidence found
- 2 in CEO Office, 1 in Server Room
- All found around 9:05 PM - 9:15 PM (after the murder)

Alibis:

- Only 4 employees provided alibis for 8:50 PM

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**INVESTIGATION
BEGINS**

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What happened in the CEO Office?

1) What evidence was found at the CEO Office?

```
SELECT  
  description  
FROM evidence  
WHERE room = "CEO Office" ;
```

description
Fingerprint on desk
Keycard swipe logs mismatch



CEO
OFFICE



KEYCARD
SWIPE LOGS
MISMATCH



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2) When was the crime discovered?

```
SELECT MIN(found_time) AS crime_discovered_at  
FROM evidence  
WHERE room = 'CEO Office';
```

crime_discovered_at
2025-10-15 21:05:00

3) What rooms have evidence?

```
SELECT DISTINCT room AS evidence_room  
FROM evidence;
```

evidence_room
CEO Office
Server Room

Key Findings to Note:

- Fingerprint on desk
- Keycard swipe logs mismatch
- Evidence discovered around 9:05 PM

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Track movements to the CEO Office

1) Who entered the CEO Office on October 15?

```
SELECT
  e.name,
  e.department,
  e.role,
  k.entry_time,
  k.exit_time
FROM keycard_logs k
JOIN employees e
ON k.employee_id = e.employee_id
WHERE k.room = 'CEO Office'
ORDER BY k.entry_time;
```



name	department	role	entry_time	exit_time
David Kumar	Engineering	DevOps Engineer	2025-10-15 20:50:00	2025-10-15 21:00:00

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2) Who was in the CEO Office during the murder window (8:30 PM - 9:30 PM)?

```
SELECT
    e.employee_id,
    e.name,
    e.department,
    k.entry_time,
    k.exit_time
FROM keycard_logs k
JOIN employees e ON k.employee_id = e.employee_id
WHERE k.room = 'CEO Office'
    AND k.entry_time <= '2025-10-15 21:30:00'
    AND k.exit_time >= '2025-10-15 20:30:00';
```

employee_id	name	department	entry_time	exit_time
4	David Kumar	Engineering	2025-10-15 20:50:00	2025-10-15 21:00:00

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3) Show ALL employee movements in the evening (after 8:00 PM)

```
SELECT
    e.name,
    e.department,
    k.room,
    k.entry_time,
    k.exit_time
FROM keycard_logs k
JOIN employees e ON k.employee_id = e.employee_id
WHERE k.entry_time >= '2025-10-15 20:00:00'
ORDER BY k.entry_time;
```

name	department	room	entry_time	exit_time
David Kumar	Engineering	CEO Office	2025-10-15 20:50:00	2025-10-15 21:00:00

Key Findings to Note:

- Only ONE person accessed CEO Office in the evening
- Timing: 8:50 PM - 9:00 PM (perfect window for 9:00 PM murder)

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Catch the liars - Who wasn't where they claimed to be?

1) Who provided alibis for 8:50 PM?

```
SELECT
    e.employee_id,
    e.name,
    a.claimed_location,
    a.claim_time
FROM alibis a
JOIN employees e
ON a.employee_id = e.employee_id
ORDER BY e.employee_id;
```

employee_id	name	claimed_location	claim_time
1	Alice Johnson	Office	2025-10-15 20:50:00
4	David Kumar	Server Room	2025-10-15 20:50:00
5	Eva Brown	Marketing Office	2025-10-15 20:50:00
6	Frank Li	Office	2025-10-15 20:50:00

CATCH THE LIARS

WHO WASN'T WHERE THEY CLAIMED TO BE?

		KEYCARD ACCESS LOGS	
DATE	TIME	WORKPLACE	REASON
15 OCT	10:32	CEO Office	Was with Debbie
20 OCT	16:19	Main Hall	Had issues
2 JAN	08:17	Main Hall	Checking mail

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2) Who has a FALSE alibi? (Claimed one place, was in another)

```
SELECT
    e.employee_id,
    e.name,
    a.claimed_location AS claimed,
    k.room AS actual_location
FROM alibis a
JOIN employees e ON a.employee_id = e.employee_id
JOIN keycard_logs k ON e.employee_id = k.employee_id
    AND a.claim_time BETWEEN k.entry_time AND k.exit_time
WHERE a.claimed_location != k.room;
```

employee_id	name	claimed	actual_location
4	David Kumar	Server Room	CEO Office

Key Findings to Note:

- Who lied about their location?
- This person is a PRIMARY SUSPECT : David Kumar
- He claimed: Server Room – but keycard shows: CEO Office.

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Who was communicating before the murder?

1) Show all calls with caller and receiver names

```
SELECT
    c.call_id,
    e1.name AS caller,
    e2.name AS receiver,
    c.call_time,
    c.duration_sec
FROM calls c
JOIN employees e1
ON c.caller_id = e1.employee_id
JOIN employees e2
ON c.receiver_id = e2.employee_id
ORDER BY c.call_time DESC;
```

call_id	caller	receiver	call_time	duration_sec
1	David Kumar	Alice Johnson	2025-10-15 20:55:00	45
5	David Kumar	Grace Tan	2025-10-15 20:40:00	90
2	Eva Brown	Alice Johnson	2025-10-15 19:30:00	120
4	Bob Smith	Jack Chen	2025-10-15 16:30:00	30
3	Clara Lee	Grace Tan	2025-10-15 14:00:00	60



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2) Who made or received calls between 8:00 PM - 9:30 PM?

```
SELECT
    e1.name AS caller,
    e2.name AS receiver,
    c.call_time,
    c.duration_sec
FROM calls c
JOIN employees e1
ON c.caller_id = e1.employee_id
JOIN employees e2
ON c.receiver_id = e2.employee_id
WHERE c.call_time BETWEEN '2025-10-15 20:00:00' AND '2025-10-15 21:30:00'
ORDER BY c.call_time;
```

caller	receiver	call_time	duration_sec
David Kumar	Grace Tan	2025-10-15 20:40:00	90
David Kumar	Alice Johnson	2025-10-15 20:55:00	45

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3) Who made the most calls during the evening?

```
SELECT
    e.employee_id,
    e.name,
    COUNT(*) AS call_count
FROM calls c
JOIN employees e ON c.caller_id = e.employee_id
WHERE c.call_time >= '2025-10-15 20:00:00'
GROUP BY e.employee_id, e.name
ORDER BY call_count DESC;
```

employee_id	name	call_count
4	David Kumar	2

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4) Did anyone make calls WHILE in the CEO Office?

```
SELECT
    e.name,
    k.room,
    c.call_time,
    e2.name AS called_person
FROM calls c
JOIN employees e ON c.caller_id = e.employee_id
JOIN employees e2 ON c.receiver_id = e2.employee_id
JOIN keycard_logs k ON e.employee_id = k.employee_id
    AND c.call_time BETWEEN k.entry_time AND k.exit_time
WHERE k.room = 'CEO Office';
```

name	room	call_time	called_person
David Kumar	CEO Office	2025-10-15 20:55:00	Alice Johnson

Key Findings to Note:

- Multiple calls by the same person (David Kumar) = Suspicious
- Calls made during CEO Office access (Alice Johnson) = Very Suspicious

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Link physical evidence to employee movements



1) Who accessed rooms where evidence was found?

```
SELECT DISTINCT
  e.employee_id,
  e.name,
  k.room,
  k.entry_time,
  k.exit_time
FROM keycard_logs k
JOIN employees e ON k.employee_id = e.employee_id
WHERE k.room IN (SELECT DISTINCT room FROM evidence)
ORDER BY e.employee_id, k.entry_time;
```

employee_id	name	room	entry_time	exit_time
4	David Kumar	Server Room	2025-10-15 08:50:00	2025-10-15 09:10:00
4	David Kumar	CEO Office	2025-10-15 20:50:00	2025-10-15 21:00:00
8	Henry Wu	Server Room	2025-10-15 08:40:00	2025-10-15 09:05:00

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2) Who accessed BOTH Server Room AND CEO Office?

```
SELECT e.employee_id, e.name
FROM employees e
WHERE EXISTS (
    SELECT 1 FROM keycard_logs k1
    WHERE k1.employee_id = e.employee_id AND k1.room = 'Server Room'
)
AND EXISTS (
    SELECT 1 FROM keycard_logs k2
    WHERE k2.employee_id = e.employee_id AND k2.room = 'CEO Office'
);
```

employee_id	name
4	David Kumar

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3) Show the complete movement timeline for suspects who accessed evidence rooms

```
SELECT
    e.name,
    k.room,
    k.entry_time,
    k.exit_time,
    CASE
        WHEN k.room IN (SELECT room FROM evidence) THEN 'EVIDENCE ROOM'
        ELSE 'Regular Room'
    END AS room_type
FROM keycard_logs k
JOIN employees e ON k.employee_id = e.employee_id
WHERE e.employee_id IN (
    SELECT DISTINCT employee_id
    FROM keycard_logs
    WHERE room IN (SELECT room FROM evidence)
)
ORDER BY e.employee_id, k.entry_time;
```

name	room	entry_time	exit_time	room_type
David Kumar	Server Room	2025-10-15 08:50:00	2025-10-15 09:10:00	EVIDENCE ROOM
David Kumar	CEO Office	2025-10-15 20:50:00	2025-10-15 21:00:00	EVIDENCE ROOM
Henry Wu	Server Room	2025-10-15 08:40:00	2025-10-15 09:05:00	EVIDENCE ROOM

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4) What does the "Keycard swipe logs mismatch" evidence mean?

```
SELECT
    e.name,
    'Keycard logs mismatch with alibi' AS evidence_connection
FROM alibis a
JOIN employees e ON a.employee_id = e.employee_id
JOIN keycard_logs k ON e.employee_id = k.employee_id
    AND a.claim_time BETWEEN k.entry_time AND k.exit_time
WHERE a.claimed_location != k.room
    AND k.room = 'CEO Office';
```

name	evidence_connection
David Kumar	Keycard logs mismatch with alibi

Key Findings to Note:

- Server Room access earlier = Possible preparation
- CEO Office access + false alibi = Direct connection to the crime
- Primary Suspect: David Kumar

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Synthesize all evidence and name the killer

1) Who meets ALL these suspicious criteria?

■ Find someone who:

1. Was in CEO Office during murder time (20:50-21:00)
2. Provided a false alibi
3. Made calls in the evening
4. Accessed Server Room earlier (possible prep)

employee_id	name
4	David Kumar

```
SELECT DISTINCT e.employee_id, e.name
FROM employees e
-- Criterion 1: In CEO Office during murder
WHERE EXISTS (
    SELECT 1 FROM keycard_logs k
    WHERE k.employee_id = e.employee_id
        AND k.room = 'CEO Office'
        AND k.entry_time <= '2025-10-15 21:00:00'
        AND k.exit_time >= '2025-10-15 20:50:00'
)
-- Criterion 2: False alibi
AND EXISTS (
    SELECT 1 FROM alibis a
    JOIN keycard_logs k ON a.employee_id = k.employee_id
        AND a.claim_time BETWEEN k.entry_time AND k.exit_time
    WHERE a.employee_id = e.employee_id
        AND a.claimed_location != k.room
)
-- Criterion 3: Made evening calls
AND EXISTS (
    SELECT 1 FROM calls c
    WHERE c.caller_id = e.employee_id
        AND c.call_time >= '2025-10-15 20:00:00'
);
```

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2) Create a complete suspect profile

```
SELECT
    e.employee_id,
    e.name,
    e.department,
    e.role,
    a.claimed_location AS alibi_claim,
    k_ceo.room AS actual_location_at_murder_time,
    k_ceo.entry_time AS ceo_office_entry,
    k_ceo.exit_time AS ceo_office_exit,
    (SELECT COUNT(*) FROM calls c WHERE c.caller_id = e.employee_id
     AND c.call_time >= '2025-10-15 20:00:00') AS evening_calls_made
FROM employees e
JOIN keycard_logs k_ceo ON e.employee_id = k_ceo.employee_id
    AND k_ceo.room = 'CEO Office'
    AND k_ceo.entry_time <= '2025-10-15 21:00:00'
    AND k_ceo.exit_time >= '2025-10-15 20:50:00'
LEFT JOIN alibis a ON e.employee_id = a.employee_id
WHERE a.claimed_location != k_ceo.room;
```

employee_id	name	department	role	alibi_claim	actual_location_at_murder_time	ceo_office_entry	ceo_office_exit	evening_calls_made
4	David Kumar	Engineering	DevOps Engineer	Server Room	CEO Office	2025-10-15 20:50:00	2025-10-15 21:00:00	2

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3) FINAL VERDICT - Name the killer!

```
SELECT name AS killer
FROM employees
WHERE employee_id = (
    SELECT DISTINCT k.employee_id
    FROM keycard_logs k
    JOIN alibis a ON k.employee_id = a.employee_id
    WHERE k.room = 'CEO Office'
        AND k.entry_time = '2025-10-15 20:50:00'
        AND k.exit_time = '2025-10-15 21:00:00'
        AND a.claim_time BETWEEN k.entry_time AND k.exit_time
        AND a.claimed_location != k.room
);
```



killer

David Kumar

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□ CASE SOLVED: CEO MURDER

Case Solved By: Gagandeep Kaur Bhatti

Killer: David Kumar

Department: Engineering

Role: DevOps Engineer

EVIDENCE AGAINST THE SUSPECT:

Physical Presence: Was in CEO Office at exact murder time (20:50-21:00)

False Alibi: Claimed to be in Server Room but was actually in CEO Office

Communication Pattern: Made 2 calls between 8:00-9:00 PM

Suspicious Access: Entered Server Room earlier in the day (possible preparation)

Evidence Match: Keycard logs mismatch directly points to this suspect

Justice delivered. Case officially concluded.



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Thank You!

For your time and attention

Special Thanks

Indian Data Club — for hosting the 21-Day SQL Challenge

DPD ZERO — for sponsoring and supporting this learning journey

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